FLEET TEAM DIGEST

737MAX-FTD-21-19002

Issue Title : Smart Ram Air Door Actuator (SRADA) Issues

Airplane Model	ATA	Minor Model(s)
737MAX	2150-00	-8, -8200, -9, -BBJ

Other Model(s)	ECCN
737NG	9E991

Originated Date	Last Revised Date	Created On
02/15/2019	09/03/2026	08/26/2026

Estimated Completion Date	Next Update Date
12/23/2027	12/18/2026

Revision Description

Added 13-May-2026 Fleet Team Meeting presentation.

Applicability

All 737 MAX airplanes

Description

Boeing has received multiple reports regarding low reliability of the Smart Ram Air Door Actuator (SRADA), P/N 67600014-1, after its first shop visit as well reports of low time Air Cycle Machine (ACM) impeller failures accompanied with exit SRADA faults. Upon investigation, ~60% of the removed SRADAs that were sent to Honeywell were determined to be No Fault Found (NFF). In response to these findings, Boeing initiated Service Related Problem (SRP) project 737NG-SRP-21-0274 for the SRADA to address two (2) issues/failure modes: a) SRADA NFF removals and b) SRADA-driven premature ACM impeller failures.

Background

The 737 MAX family of airplanes has carried forward the high-pressure water separation pack and the SRADA P/N 67600014-1 system on all minor models 737-8/-9/-8200. The 737 MAX BIT interface and system fault reporting from the Integrated Air System Controller (IASC) is via the flight deck MAX display system. With this evolution of the air conditioning system BIT and fault reporting, the LEDs are redundant and no longer required for the 737 MAX airplanes.

In an effort to understand the high percentage of SRADA NFF/NCFR removals, Boeing and Honeywell have completed the following tasks:

- Revised various 737 MAX IFIM tasks to address power sequencing issues to prevent nuisance faults stored within the SRADA internal fault memory. Currently, there is an IASC software change in work to eliminate SRADA nuisance faults by implementing fault confirmation time improvements.

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- During the testing of returned units, the team witnessed a potentiometer (i.e., position sensor) related fault at the Exit SRADA position when commanded to Cruise mode. Honeywell's Repair & Overhaul (R&O) shop developed a potentiometer noise test for troubleshooting units that fail ATP.
- Conducted performance and high-temperature testing of the control boards and/or interface PBA.
- During testing, approximately 2% of circuit card failures were found to have delamination within the R9 resistor.
- Incorporated AMM/IFIM improvements with instructions to check for mechanical wear/damage, replace worn components if found (in lieu of replacing the SRADA itself), as well as provide supplemental tools to aid in Exit SRADA rigging installation.
- Worked with operators to inspect the mechanical condition of the Exit SRADA extendable end arms and pins.
- Performed ground tests with previously faulted SRADAs on a 737 MAX production airplane and a 737NG in-service airplane using a custom wire harness to connect the Inlet and Exit SRADA to Honeywell's UNITY data recorder. The intent of this effort was to collect Non-Volatile Memory (NVM) data while reproducing faults seen in-service. There were no findings during this effort.
- Performed lab testing (i.e., vibration, temperature cycling, Electro-Magnetic Interference). There were no findings during this effort.
- Performed SRADA actuator-level power quality and power interruption testing. There were no findings during this effort.
- Developed software to filter through PFTC NVM data to correlate aircraft operating conditions with SRADA removals. This effort was completed in October of 2021. There were no findings during this effort.
- Performed an on-wing dimensional analysis fleet survey to determine if tolerance stack-up between the inlet linkage assembly and inlet SRADA could cause potential for bottoming out. Fleet measurements show the potential for bottoming out in the ground full open position. As result of these findings, Boeing procured a ram air inlet assembly to integrate into our SIL; however, due to the recent findings discussed below, this effort is on pause.
- Reviewed AHM fleet data for SRADA fault to determine if location (inlet/exit/left/right) or flight phase could be contributing to SRADA faults. No conclusions were made.

Note: A new SRADA, P/N 67600046-1, was introduced into the 737 MAX fleet. For more information, please refer to Service Letter 737-SL-21-157. The new SRADA P/N 67600046-1 does not address the NFF / NCFR issues discussed in this FTD article.

Status

Boeing is continuing to investigate SRADA NFF removals and ACM impeller failures by looking into the following areas of interest:

1. System Integration Lab (SIL) Testing: Utilize our SIL, which includes two Integrated Air System Controllers (IASC), two Air Conditioning Accessory Units (ACAU), and 4 SRADAs that are instrumented with physical and commanded position recording equipment. Boeing can simulate flight profiles that will actuate the SRADAs while recording the physical and commanded positional data along with lower-level Continuous Built-In Test (CBIT) monitoring. This effort is intended to determine the SRADA's control accuracy and record lower-level CBIT information that is not available on the aircraft. The goal is to duplicate the failures seen on-wing and determine what SRADA sub-components are causing the faults.
2. Honeywell Shop Efforts: In coordination with Boeing's SIL testing, Honeywell is using the data learned from lab testing to improve their Component Maintenance Manual (CMM) level incoming test to better catch degraded SRADA sub-components. Additionally, Honeywell is using historical shop data to determine the survivability of SRADAs after their first shop visit based on what components were replaced.

Reference the latest Working Together Team (WTT) meeting presentation from 21-Oct-2025 and 13-May-2026 for further details regarding the efforts noted above.

Lastly, Boeing is working on a design change that will add wiring from the ram air temperature sensors to the Digital Flight Data Acquisition Unit (DFDAU). This change will enable the DFDAU to capture and record the ram air sensor temperature reading and make it available in ADR-QAR output. Utilizing this data will help troubleshoot potential SRADA anomalies before they fail, which

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may help prevent premature ACM impeller failures. This change is expected to be introduced into 737 MAX production first and will require a wiring change and DFDAU software change. Following 737 MAX production incorporation, a Service Bulletin will be issued for delivered 737 MAX.

Interim Action

Boeing recommends operators to follow the associated Fault Isolation Manual (FIM) task for any active Maintenance Messages (MM).

To ensure that the inlet and exit linkage assembly are operational, it is recommended to periodically perform the following, especially if you are experiencing chronic SRADA faults on the same aircraft and same installed location:

1. Inlet Linkage Assembly: AMM TASK 21-51-22-200-801 “Ram Air Inlet Door Linkage Inspection”
2. Exit Linkage Assembly: With the exit SRADA removed, ensure smooth rotation of the link arm throughout its entire stroke. Reference Boeing CMM 21-51-03 for any repairs needed to the exit assembly.

Final Action

To be determined, pending SRADA NFF investigation.

Milestones

Honeywell SRADA NFF Program Kick-off: 28-SEP-2018 (complete)
AMM Update (Inspect SRADA linkage mechanisms for wear): 15-OCT-2019 (complete)
AMM Update (Use force gage to check resistance of SRADA linkage): 15-OCT-2019 (complete)
Boeing / Honeywell Ground Test on 737MAX production airplane: 6-OCT-2020 (complete)
Boeing / Honeywell Ground Test on in-service airplane (in coordination with participating operators): 29-JAN-2021 (complete)
SRADA Power Quality Testing: AUG-2021 (complete)
Integrated Systems Lab SRADA Testing Setup: MAR-2024 (complete)
On-Wing Dimensional Measurement/Analysis Knowledge Point (In-Service and Production Aircraft): 14-JUN-2024 (complete)
DFDAU B.P 1.5 737-SL-31-094 Release: 27-JAN-2026 (complete)
Ram Air Sensor Wiring-Production Incorporation Skip Tab: Q1 2027 (target)
Ram Air Sensor Wiring-Full Production Incorporation: Q2 2027 (target)
Ram Air Sensor Wiring Change - 737 MAX SB: Q3 2027 (target)
Systems Integration Lab (SIL) SRADA Testing Data Collection: On-Going Effort

Operator Action

Based on the findings from the recent SIL testing and Honeywell shop data review, Honeywell recommends operators to ensure that the potentiometer gets replaced even if testing No Fault Found (NFF) during the CMM incoming test if the SRADA has more than 5 years of service (roughly 15,000 - 20,000FHs). Note that if you send your SRADAs to Honeywell for repair and overhaul, Honeywell will recommend this as part of a new internal process. Please reference Honeywell Service Information Letter (SIL) D202604005260R000 for more information.

Reference Categories

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Reference Type	Reference Number	Revision Number	Reference Date
Service Related Problem (SRP)	737NG-SRP-21-0274		
Service Letter (SL)	737-SL-21-157		

Parts List

Part Name	Boeing Part Number	Supplier Name	Supplier Number
Smart Ram Air Door Actuator (SRADA)		Honeywell	67600046-1
Smart Ram Air Door Actuator (SRADA)		Honeywell	67600014-1
Integrated Air Systems Controller (IASC)		Honeywell	51090322-002

Related Categories

- Airplane Maintenance Issues : Issues that occur during airplane maintenance or have an impact on airplane reliability
- Component Issues : Issues that occur during component maintenance or have an impact on component reliability

Attachments:

a) SRADA WTT Presentation May 2026

b) SRADA WTT Presentation October-2025

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